

# AK VisionFlow — Phase 9

Claude Development Prompt  
AI + Business Intelligence + Predictive Analytics

**Critical context:** Phase 1–8 provide the core POS, inventory, ERP/accounting, procurement, multi-branch, warehouse, optical, clinic, patient, prescription, laboratory, CRM, WhatsApp, automation and customer-portal foundation. Phase 9 is an incremental intelligence layer. Inspect the existing repository first. Do not rebuild working functionality, reset the database, or delete/corrupt existing tenant data.

## 1. Mission

Transform AK VisionFlow from an integrated business management platform into an AI-assisted, data-driven Business Operating System. AI must provide practical business intelligence, forecasting, anomaly detection, recommendations and natural-language analysis using the tenant's authorized data. AI must assist decision-making, not silently perform high-risk financial, clinical or operational actions.

## 2. Core AI Philosophy

**Do not build AI for marketing language. Build AI that saves the owner time, reduces losses, improves decisions and increases revenue.**

Every AI insight should be explainable enough for a business user: what changed, why it matters, what data supports it, what action is recommended, and how confident/uncertain the system is.

## 3. Non-Negotiable Rules

- Preserve all existing Phase 1–8 functionality.
- Never reset, truncate, recreate or destructively migrate the production database.
- All schema changes must use safe, versioned migrations.
- Existing tenants, users, customers, patients, products, inventory, sales, purchases, accounting, branches, warehouses, optical orders, prescriptions, lab jobs, messages and portal records must remain intact.
- AI must enforce tenant isolation and existing RBAC.
- AI must only retrieve data the requesting user is authorized to access.
- Never send another tenant's data into an AI context.
- Do not expose sensitive clinical/patient data to unauthorized roles.
- Do not let AI directly post accounting entries, change stock, alter prescriptions, approve purchases, issue refunds or perform other high-impact actions without explicit authorized confirmation and deterministic backend validation.
- Financial calculations and source-of-truth reports remain deterministic/backend-authoritative; AI may interpret them but must not replace them.
- AI-generated recommendations must be clearly distinguished from factual system data.
- Use confidence/uncertainty indicators where predictions are involved.
- External AI providers must be configurable and credentials must remain server-side.
- Do not train a general model on one tenant's private data for another tenant.
- Run the complete existing test suite and add comprehensive Phase 9 tests.

## 4. AI Business Assistant

Create an owner/admin-facing AI Business Assistant integrated into the Advanced Business Command Center.

- Natural-language business questions.

- Date-aware answers.
- Branch-aware answers.
- Product/category-aware answers.
- Customer/supplier-aware answers subject to permissions.
- Sales/profit analysis.
- Inventory analysis.
- Receivables/payables analysis.
- Procurement analysis.
- Optical/clinic operational analysis.
- Trend comparison.
- Explainable recommendations.
- Links/drill-downs to the underlying records and reports.

## 5. Natural-Language Reporting

- “Show me this month's sales compared with last month.”
- “Which branch is most profitable?”
- “Which products have the best gross margin?”
- “Which customers have overdue payments over 60 days?”
- “Which products may run out soon?”
- “What are my slow-moving products?”
- “Why did profit decline this month?”
- “Show me pending optical jobs older than the expected delivery date.”
- “How much stock is tied up in slow-moving products?”
- “Which supplier prices increased the most?”
- “What should I pay attention to today?”

## 6. AI Daily Business Brief

Generate a concise daily management brief for authorized owners/admins using deterministic business metrics plus AI interpretation.

- Sales performance.
- Profit trend.
- Cash/receivables/payables.
- Inventory risks.
- Expiry risks.
- Low-stock/reorder risks.
- Pending/delayed optical jobs.
- Appointment/clinic activity.
- Important approvals.
- Unusual activity.
- Top three recommended actions.

- Links to relevant reports.

## **7. Sales Forecasting**

- Forecast future sales using historical transaction data.
- Support daily/weekly/monthly forecasting.
- Allow branch/category/product-level forecasting where data volume is sufficient.
- Show historical vs forecast values.
- Show confidence ranges/uncertainty.
- Detect seasonal patterns where sufficient history exists.
- Clearly label forecasts as estimates, not guaranteed outcomes.
- Gracefully handle insufficient or poor-quality data.

## **8. Inventory Demand Intelligence**

- Demand forecasting.
- Reorder recommendations.
- Suggested reorder quantity.
- Expected days of stock remaining.
- Low-stock risk prediction.
- Overstock detection.
- Slow-moving stock.
- Dead-stock identification.
- Stock aging.
- Branch-specific demand.
- Central replenishment recommendations.
- Medicine expiry risk.
- Capital tied up in slow-moving inventory.

## **9. Product & Margin Intelligence**

- Best-selling products.
- Most profitable products.
- Low-margin products.
- Products with declining sales.
- Products with unusual price/cost changes.
- Category profitability.
- Supplier price impact.
- Discount impact on margin.
- Branch-specific product performance.
- Recommended products/categories for management attention.

## **10. Customer Intelligence**

- New vs returning customers.
- Customer purchase frequency.
- Customer lifetime-value indicators.
- Inactive customer detection.
- Potential churn indicators.
- High-value customer identification.
- Outstanding-payment risk indicators.
- Follow-up recommendations.
- Customer segment insights.
- Campaign opportunity suggestions for Phase 8 automation.

## **11. Optical & Clinic Intelligence**

- Optical order volume trends.
- Pending/delayed job analysis.
- Lab turnaround-time trends.
- Lab performance comparison.
- Examination-to-optical-order conversion.
- New vs returning patient/customer trends.
- Appointment/no-show trends.
- Follow-up opportunities.
- Branch/doctor operational trends.
- Do not generate clinical diagnoses or treatment recommendations as a substitute for qualified clinicians.

## **12. Anomaly & Risk Detection**

- Unusually high discounts.
- Abnormal returns/reversals.
- Unexpected stock adjustments.
- Unusual expense spikes.
- Unexpected sales decline.
- Unusual purchase price increases.
- Abnormal cash differences where data supports it.
- Sudden branch performance changes.
- Unusual customer-payment behavior.
- Unusual optical-job delays.
- Potential duplicate/abnormal transactions.

## **13. AI Recommendation Center**

Create a prioritized list of recommendations with severity/priority, evidence, expected impact and drill-down.

- Urgent: action may prevent loss.
- Attention: meaningful business issue.

- Opportunity: potential revenue/profit improvement.
- Information: useful trend with no immediate action.
- Every recommendation must link to the underlying data/report.
- Allow authorized users to dismiss/acknowledge recommendations.
- Store recommendation history so repeated noise can be reduced.

## **14. AI in the Advanced Command Center**

- AI Business Summary card.
- Top business risks.
- Top opportunities.
- Forecast cards.
- Inventory recommendations.
- Customer follow-up opportunities.
- Anomaly alerts.
- Branch performance explanations.
- Natural-language query box.
- Drill-down from AI insight to deterministic report/source records.

## **15. Explainability & Trust**

- Show the period/data scope used for an insight.
- Show the key supporting metrics.
- Link to the source report or records.
- Clearly distinguish facts from predictions/recommendations.
- Show confidence or uncertainty for predictive outputs.
- Avoid fabricated numbers or unsupported claims.
- If the data is insufficient, say so instead of guessing.
- Provide a feedback mechanism for incorrect/unhelpful insights.

## **16. AI Data & Privacy Architecture**

- Use tenant-scoped data access at the service layer.
- Enforce user/role/branch permissions before constructing AI context.
- Minimize data sent to external AI providers.
- Do not send unnecessary personal/clinical information.
- Mask/redact sensitive fields where possible.
- Keep provider API keys server-side.
- Log AI requests safely without storing unnecessary sensitive content.
- Define retention for AI interaction history.
- Provide tenant-level AI enable/disable/configuration controls.
- Prevent cross-tenant prompt/context contamination.

## 17. AI Provider Architecture

- Use a provider abstraction layer.
- Allow configuration of the supported AI provider/model.
- Keep application business logic independent of one provider.
- Implement timeouts, retries and graceful failure.
- Provide deterministic fallback to normal reports when AI is unavailable.
- Track AI usage/cost where the provider exposes usage data.
- Apply tenant-level usage limits/quotas where appropriate.
- Do not block core business operations because an AI provider is unavailable.

## 18. AI Actions — Human Confirmation Required

- AI may recommend a purchase; authorized user must approve before PO creation.
- AI may recommend stock reorder; authorized user must confirm before creating a procurement transaction.
- AI may identify overdue customers; automation may be triggered only through configured Phase 8 workflows.
- AI must not directly change accounting balances.
- AI must not directly change inventory quantities.
- AI must not directly modify prescriptions or clinical records.
- AI must not approve expenses, discounts, refunds or purchases without explicit authorized confirmation.
- All confirmed actions must pass normal backend validation, permissions and audit controls.

## 19. AI Usage & Cost Controls

- Track AI requests and approximate usage/cost.
- Set sensible limits per tenant/user.
- Cache safe repeated analytical results where appropriate.
- Prefer deterministic database aggregation before invoking AI.
- Do not send raw entire database contents to an AI model.
- Use summarized/structured data for AI context.
- Prevent runaway loops or repeated automation-triggered AI calls.

## 20. Performance & Reliability

- AI must not slow normal POS, sales, inventory, accounting or clinic operations.
- Use asynchronous processing for heavy forecasting/analytics.
- Provide loading/progress states.
- Cache or precompute suitable aggregates.
- Handle AI-provider downtime gracefully.
- Do not make core business transactions dependent on an AI response.
- Keep dashboards responsive even when AI insights are unavailable.

## 21. Reporting & Analytics

- Forecast accuracy monitoring.

- AI recommendation history.
- Anomaly detection history.
- AI usage/cost report.
- Most-used AI questions.
- AI insight acceptance/dismissal feedback.
- Inventory forecast vs actual demand.
- Sales forecast vs actual sales.
- Branch trend analysis.
- Customer retention trends.

## 22. UX Requirements

- Make AI accessible from the Owner/Admin Command Center.
- Use simple business language rather than technical AI terminology.
- Show source metrics and drill-down links.
- Clearly label predictions and recommendations.
- Allow the user to ask follow-up questions.
- Provide useful suggested questions/prompts.
- Do not force AI usage; normal reports must remain available.
- Maintain existing responsive/PWA/light-dark UI.

## 23. Testing Requirements

- Test tenant isolation of AI context and queries.
- Test branch/RBAC filtering before AI context creation.
- Test that unauthorized customer/patient/clinical data cannot enter AI context.
- Test deterministic source numbers against known fixtures.
- Test AI answer grounding to source metrics.
- Test insufficient-data behavior.
- Test provider timeout/failure and deterministic fallback.
- Test idempotency for AI-triggered workflows.
- Test usage/cost limits.
- Test that AI cannot bypass approval/accounting/inventory controls.
- Test forecast output with controlled historical data.
- Test anomaly detection against known abnormal scenarios.
- Run the complete regression suite from previous phases.

## 24. Migration & Data Safety

Inspect the existing database and Command Center before adding AI tables. Add only the required entities for AI conversations, insight history, forecasts, anomaly events, recommendation state, feedback, provider configuration and usage tracking. Do not duplicate core business data. Existing records must remain untouched.

## 25. Required Deliverables

- AI Business Assistant.
- Natural-language business reporting.
- Daily AI Business Brief.
- Sales forecasting.
- Inventory demand/reorder intelligence.
- Product and margin intelligence.
- Customer intelligence.
- Optical/clinic operational intelligence with safe boundaries.
- Anomaly/risk detection.
- AI Recommendation Center.
- Command Center AI integration.
- Explainability/source drill-down.
- Tenant/RBAC-aware AI privacy architecture.
- Provider abstraction and AI failure fallback.
- AI usage/cost controls.
- Safe migrations.
- Automated tests and regression results.
- Technical documentation and Phase 9 completion report.

## 26. Acceptance Criteria

- Existing Phase 1–8 functionality continues to work.
- No existing data is deleted, reset or corrupted.
- AI answers are scoped to the authorized tenant, branch and user.
- Core factual numbers come from deterministic system data.
- AI clearly distinguishes facts, predictions and recommendations.
- Natural-language business questions can be answered with useful drill-downs.
- Daily business brief provides actionable owner insights.
- Sales and inventory forecasting works where sufficient historical data exists.
- Slow-moving/dead stock and reorder intelligence is useful and explainable.
- Anomaly detection identifies configured/high-value abnormal patterns.
- AI cannot bypass accounting, inventory, clinical or approval controls.
- Provider failure does not interrupt core business operations.
- AI usage/cost controls work.
- All existing tests plus new Phase 9 tests pass.
- No critical security, privacy, data-integrity or AI-safety issue remains open.

## 27. Claude Execution Order

1. Inspect the existing repository, Command Center, accounting, inventory, CRM, clinic, optical, automation and portal architecture.
2. Run the complete existing test suite and establish a baseline.



3. Define the deterministic data/analytics layer that AI will consume.
4. Design tenant/RBAC-aware AI context and privacy architecture.
5. Design safe migrations for AI configuration, insight and usage data.
6. Implement provider abstraction and secure AI configuration.
7. Implement natural-language business query service with deterministic data retrieval.
8. Implement AI Business Assistant and source drill-down.
9. Implement Daily Business Brief.
10. Implement sales forecasting.
11. Implement inventory demand/reorder intelligence.
12. Implement customer/product/margin intelligence.
13. Implement anomaly detection.
14. Implement recommendation center and feedback.
15. Integrate AI into the Advanced Command Center.
16. Implement usage/cost controls and graceful AI failure fallback.
17. Add comprehensive security, privacy, grounding and regression tests.
18. Run full regression and production-readiness review.
19. Document architecture, data boundaries, provider setup, limitations and operational procedures.

## Final Instruction to Claude

**Build Phase 9 as a safe, production-quality intelligence layer on top of the existing AK VisionFlow platform. Do not rebuild the core and do not reset the database. AI must be genuinely useful to business owners, not a marketing label. Use deterministic application data as the source of truth and use AI for interpretation, explanation, prediction and recommendations. Every AI result must respect tenant isolation, RBAC, branch scope and privacy boundaries. Never allow AI to silently perform high-impact financial, inventory or clinical actions. Require explicit authorized confirmation and normal backend validation for consequential actions. The system must remain fully functional if the AI provider is unavailable. The final result should make AK VisionFlow an intelligent Business Operating System for optical, eye-care and medical businesses, with actionable insights available directly from the Owner/Admin Command Center.**